

Process Simulation CHE 433 Assignment 1

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1 Complete Process Flow

Figure A: Hydrogenation of Benzene to Cyclohexane

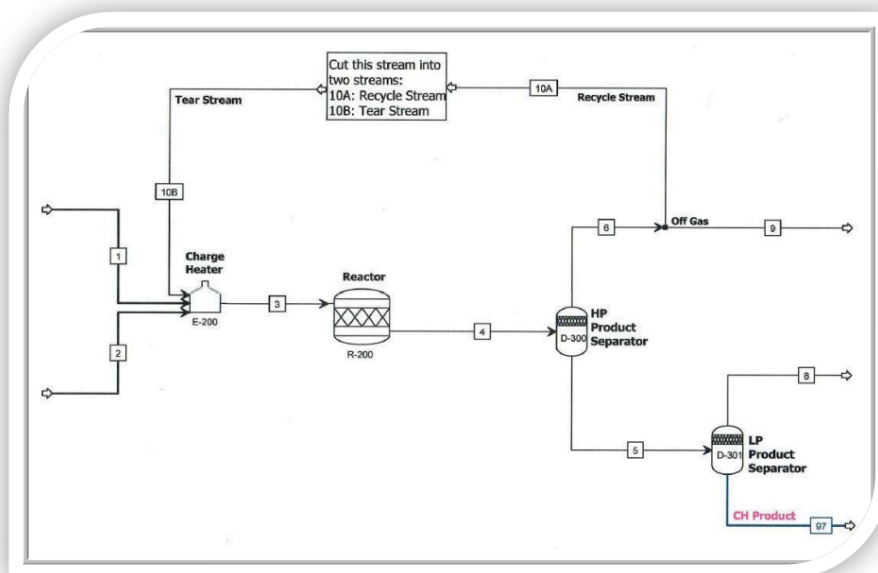


Figure 1: Original Process Flow

As we were looking at the process flow sheet and the given information we noticed a few unit operations that needed to be added in for the process flow to work properly. Starting off with stream 1 the benzene feed is entering into the charge heater at 15 psia while stream 2 the Hydrogen feed is entering at 335 psia. We want both streams to be equal in pressure so we add in a pump to stream 1 to increase pressure from 15psia to 335 psia.

We do not want to add all the streams 1,2, and 10B straight into the charge heater. Before doing so we need to add in a mixer to properly mix the three streams together.

The next unit operation that we need to add in is in-between the reactor and High Pressure (HP) separator. The temperature exiting the reactor is at $400^{\circ}F$ and the temperature at the HP flash drum is at $120^{\circ}F$. We need a unit operation here to drop the temperature. So we added in a condenser to help reduce the temperature.

In between the recycle stream and the tear stream adding a compressor will help with increasing the temperature and pressure. As the temperature and pressure exiting the HP flash drum are 300psia and $120^{\circ}F$. Using the compressor will help increase it to the desired 335psia to enter the mixer at the same pressure as stream 1 and 2.

The last unit operator that we added is at the bottom stream at the HP flash drum. To decrease the pressure from 300psia to 15 psia we will add in a pressure regulator.

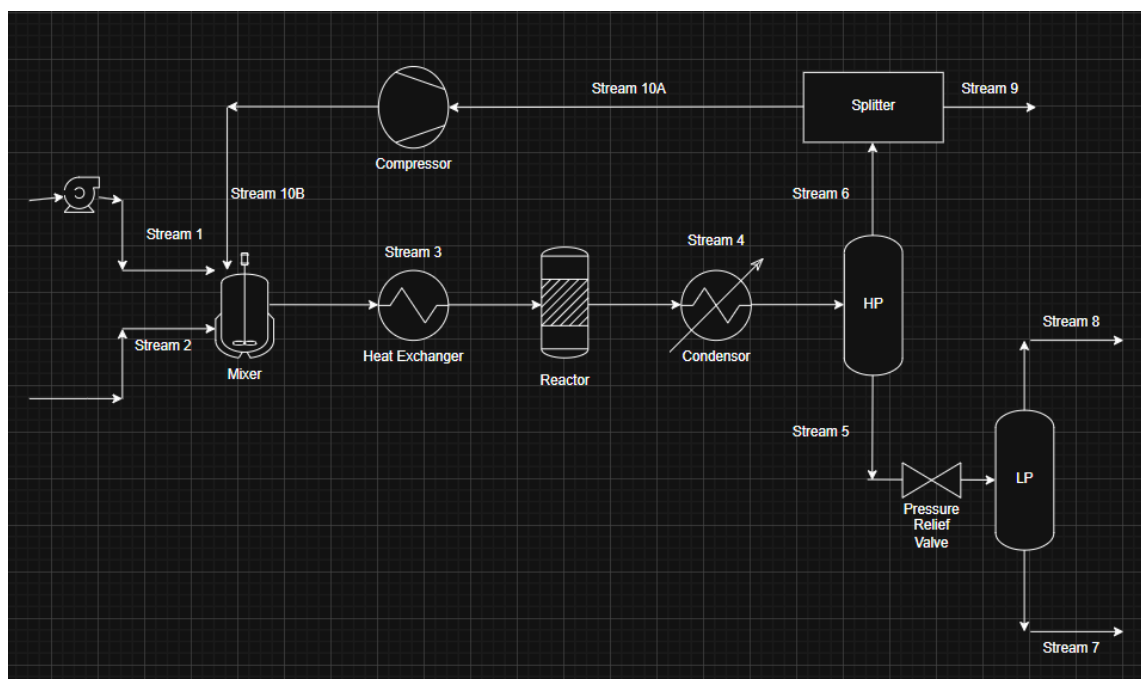


Figure 2: New Process Flow Design

2 Materials Balance

To start of the material balance we need to get all our information down. We will start off by solving what is coming out from all the unit operations and save the recycle stream for the end as that will make it easier for us to calculate later on.

ENG Calculations							
Molar Composition	I	II	10B	MIXER	HEATER	III	RX
C6H6	1.0000	0.0000		0.2451	0.2451	0.2451	0.2451
H2	0.0000	0.9750		0.7360	0.7360	0.7360	0.7360
N2	0.0000	0.0050		0.0038	0.0038	0.0038	0.0038
CH4	0.0000	0.0200		0.0151	0.0151	0.0151	0.0151
C6H12	0.0000	0.0000		0.0000	0.0000	0.0000	0.0000
TOTAL	1.0000	1.0000	0.0000	1.0000	1.0000	1.0000	1.0000
Stream Properties	I	II	10B	MIXER	HEATER	III	RX
Molar Flow, lbmol/hr	92.1400	283.8000		375.9400	375.9400	375.9400	375.9400
Mass Flow, kg/s							
Volume Flow, m ³ /s							
Temperature, °C	37.7778	48.8889			148.8889		204.4444
Pressure, bar	1.0342	23.0974			22.7526		21.7184

Figure 3: Streams Before Reaction

To begin we have 2 starting streams. Stream 1 consists of Benzene. With a flow of 92.14 lbmol/hr at 100°F and 15psia. Hence why the pump is needed to increase the pressure to 335 psia. Stream two consists of Hydrogen with the following mixture: 0.975 H_2 , 0.005 N_2 , and 0.020 CH_4 . The Feed comes in at 283.8 lbmol/hr. Since the next unit operator is a mixer the only thing that will change is the molar flow rate and the molar composition of the mixture. To find the molar flow we simply add the two incoming streams together to give us a total of 375.96 lbmol/hr. To get the molar composition of the mixture we take the flow rate of stream 1 and divide it by the total flow rate which will give us a molar composition of 0.245 benzene. Since stream 2 is made up of 3 different elements we find the molar composition a little differently. Take mole comp of hydrogen and multiply it by the flow rate of stream 2 and divide it by the total flow rate coming out of the mixer. Repeat this with Nitrogen and Methane which will give us the following molar composition of the mixture. Benzene: 0.2451, H_2 : 0.7360, N_2 : 0.0038, CH_4 : 0.0151. The molar composition will stay the same in the heat exchanger and will only change once leaving the reactor.

				Conversion	0.9986	
Molar flows into Reactor (lbmol/hr)	Stream III			OUT	Stream IV	Compositions
C6H6	92.1402			C6H6	0.1290	0.0012
H2	277.3084			H2	1.2748	0.0122
N2	2.4957			N2	2.4957	0.0238
CH4	8.7611			CH4	8.7611	0.0837
C6H12	0.1417			C6H12	92.0112	0.8790
Total	380.8471			Total	104.6718	1.0000

Figure 4: FReactor Calculations

In the problem statement, we are given the conversion rate of the reactor being 99.86 percent. Using this and the following equations we can find the molar composition leaving the reactor. Using the molar composition previously mentioned in the reactor we can find the molar flow of each element.

We do this by multiplying the molar composition of the specific element and the total molar flow. Benzene: $0.2451 * 375.94 = 92.14 \text{ lbmol/hr}$ repeat this for the rest of the molar flows.

Now that we have the molar flows we need to find the flow rate exiting the reactor as well as the compositions. Using the following formulas we can calculate the flow rates.

$$F_{bz} = F_{bz}^{\circ} - (F_{bz}^{\circ} * \text{conversion})$$

$$F_{C_6H_{12}} = F_{bz}^{\circ} * \text{conversion}$$

$$F_{H_2} = F_{H_2} - F_{C_6H_{12}}$$

Nitrogen and Methane are both inert gases, so their molar compositions will stay the same. Using the equations listed we can solve for the flow rate and molar compositions. Which gives us the following

Molar flow rates and compositions:

Benzene: 0.129 lbmol/hr and 0.0005

Hydrogen: 184.69 lbmol/hr and 0.6505

Nitrogen: 1.419 lbmol/hr and 0.005

Methane: 5.676 lbmol/hr and 0.02

CycloHexane: 92.011 lbmol/hr and 0.3242

		OUT					
In/Out HP Sep. (lbmol/hr)	zi	xi	yi	ki	RR		Alpha
C6H6	0.0005	0.0013	0.0000	0.0217	0.0013		0.6748
H2	0.0986	0.0012	0.1455	117.5000	-0.1442		
N2	0.1636	0.0057	0.2396	42.0000	-0.2339		
CH4	0.4138	0.0374	0.5952	15.9000	-0.5578		
C6H12	0.3236	0.9543	0.0197	0.0206	0.9346		
Totals	1.0000	1.0000	1.0000	175.4423	0.0000	0.0000	0.0000
		OUT					
In/Out LP Sep. (lbmol/hr)	zi	xi	yi	ki	RR		Alpha
C6H6	0.0013	0.0014	0.0004	0.3214	0.0009		0.0607
H2	0.0012	0.0000	0.0203	2461.0000	-0.0203		
N2	0.0057	0.0001	0.0923	842.0000	-0.0922		
CH4	0.0374	0.0019	0.5870	306.0000	-0.5851		
C6H12	0.9543	0.9966	0.3000	0.3010	0.6966		
Totals	1.0000	1.0000	1.0000	3609.6224	0.0000		

Figure 5: HP and LP calculations

The molar comp entering the condenser will be the same as that of the reactor and will only change once leaving the high-pressure (HP) separator as the mixture is separated by liquid and vapor. Zi is our mol composition for each element. Ki are the values that Dr. Caracotsios provided us. We needed to solve for Xi, Rachford-Rice, and Alpha (Vapor-Liquid Fraction). First, we need to have an initial guess for alpha. Just any reasonable number will do as we will then proceed to use the "Goal-Seek" method in which we set our RR to 0 via changing the Alpha value. Now we use the following equations to find the RR and the liquid vapor fraction. For each of the elements. You want to repeat these steps with the low pressure separator. However, the Zi values come from the HP separator calculation Xi values. Repeat the calculations from the HP separator

Rachford-Rice

$$\frac{Z_i(K_i - 1)}{1 + \alpha(K_i - 1)} = 0$$

Liquid phase

$$\frac{Z_i}{1 + \alpha(K_i - 1)}$$

Vapor Phase

$$K_i * X_i$$

Molar Composition	10 RECYC	10 TEAR		
C6H6	0.0000	0.0000		0.0002
H2	0.1388	0.0867		1.1610
N2	0.2341	0.1547		1.9588
CH4	0.6074	0.4432		5.0820
C6H12	0.0196	0.0204		0.1644
TOTAL	1.0000	0.7050		

Figure 6: Recycle and Tears Stream

The last part is now to solve for the recycle stream. We understand that the Vapour coming out of the HP Separator are the "xi" values we obtained from the previous calculations and is what travels to the splitter and "10A" stream. To start our calculations for the recycle stream, we will create two columns, one for the recycle and one for the tear. Here the tear stream column will be given an initial guess value which we will then update via the "Newton Method" by selecting all the compositions from the recycle column, total flow of the column and the RR, all on the LHS. On the RHS we select the same from the tear column and we include the Alpha value instead of the RR, we do this to change the same cells in the tear stream.

10 RECYC	10 TEAR
0.0000	0.0000
0.1455	0.1455
0.2396	0.2396
0.5952	0.5952
0.0197	0.0197
1.0000	1.0000
10 RECYC	10 RECYC
188.1823	188.1823

Figure 7: Recycle & Tear Streams

Using the previously mentioned method, we see that our recycle and tear streams converge at this point. Therefore, we can proceed with all other calculations regarding costs. Our final streams entering the mixer will then be:

I	II	10B
1.0000	0.0000	0.0000
0.0000	0.9750	0.1455
0.0000	0.0050	0.2396
0.0000	0.0200	0.5952
0.0000	0.0000	0.0197
1.0000	1.0000	1.0000
I	II	10B
92.1400	283.8000	188.1823

Figure 8: Final flows

3 Cost of Utilities

For the cost evaluation, we were provided with the numbers:

Cooling water : 0.345\$/GJ

Natural Gas: 6.00\$/GJ

Electricity: 0.06\$/kWh

Here all we must do is simply calculate the power each unit operation takes.

ENERGY	ELECTRIC		ELECTRIC		GAS		WATER		ELECTRIC	
STREAM	PUMP IN	PUMP OUT	COMP IN	COMP OUT	HEAT IN	HEAT OUT	HE IN	HE OUT	VALVE IN	VALVE OUT
TEMP (C)	37.8	38.9863	48.8889	50.1494	45.7900	148.8889	204.4444	48.8889	48.3333	44.5274
PRESS (bar)	1.0342	23.0974	22.7526	23.0974	23.0974	22.7526	21.7184	21.7184	22.7526	1.0342
FLOW (kmol/s)	0.0116	0.0116	0.0237	0.0237	0.0711	0.0711	0.0358	0.0358	0.0117	0.0117
Z	0.0035	0.0786	0.9759	0.9759	0.9503	0.9823	0.9520	0.8136	0.0882	0.0041
ENTHALPY (kJ/s)	51836.12	52113.12	-46440.069	-46397.520	-2478.7542	2227.2673	-71258.5108	-86384.153	-147580.819	-148286.0555
POWER (kW/h)	3.22		1.0089		4706.0214		15125.6425		8.2169	

Figure 9: Power Calculations

Here, we have fully calculated the total power costs needed for the unit operations in kWh. Using the given temperatures & pressures, calculating new ones needed, we proceeded to find the molar flow for each operation in kmol/s along with each compressibility factor, both for in & out. Using these values and the listed prices of each utility, we then finally have the cost we need for the whole operation.

COST EVALUATION	\$					
Cooling Water	0.3450	\$/GJ				
Natural Gas	6.0000	\$/GJ				
Electricity	0.0600	\$/kWh				
	Power (kWh)	Power (GJ)	COST (\$)		TOTAL COSTS	68959.3346
Pump	0.0009	0.0000	1691.3203			
Compressor	0.0003	0.0000	530.6153			
Heater	4706.0214	3.7648	54213.3670			
Condenser	15125.6425	12.1005	12524.0320			

Figure 10: Cost Evaluation

The costs of each utility in the process has now been evaluated and as we can see, it costs \$68,000/yr to run the operation.

4 Reduction of costs

Looking at our calculations the main costs come from the use of the heater and condenser. Which uses water and gas. One of the ways to reduce the cost is replacing the heater from being gas to an electric heater. Depending on the type of electric heater we go with it can be more efficient and allows us to have more accurate control of the temperature. Lowering the cost per year significantly.

COST EVALUATION	\$					
Cooling Water	0.3450	\$/GJ				
Natural Gas	6.0000	\$/GJ				
Electricity	0.0600	\$/kWh				
	Power (kWh)	Power (GJ)	COST (\$)		TOTAL COSTS	\$/yr
Pump	0.0009	0.0000	1691.3203			
Compressor	0.0003	0.0000	530.6153			
Heater	4706.0214	3.7648	316.2446			
Condenser	15125.6425	12.1005	26717.9350			

Figure 11: Cost Per Year with Electrical Heater

Another idea could be use steam/water to keep the stream before the heater warmer so that we use less power in heating up the stream to the temperature desired.